AI-Powered Algorithmic Trading Among Individual Investors: A Research Mapping

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Abstract

AI based Algo trading has played a significant role by enabling more efficient, faster, and data-driven decision-making processes in financial markets. There have been plethora of researches conducted on Algo trading , and there is no controversy on AI-driven trading creates more efficient and liquid markets, but how receptive the retail investor was the primary objective of the study .

While reviewing the existing literature, it was found that as many as 2059 papers were on AI , Algo Trading and Retail Investors. This led the researchers to do Bibliometric analysis based on relevant keywords, applied restrictions and thresholds, and thus 189 research papers were filtered.

Terms were searched on WOS -"Algorithm trading" OR "Automated Trading" OR "Black-Box trading" OR "Algo- trading" AND "Artificial Intelligence" OR "Role of Artificial Intelligence in Algo Trading" AND "Retail-investors", 182 papers were generated from WOS, and after limiting the papers to article, proceeding paper, review article, language English and using PRISMA framework the final papers were 178, which were 2,489 times cited with 27 h-index. Further PRISMA framework is also used in which 8 papers from other sources were also taken.

Analysis was done with the help of "Vosviewer" software and WOS database. Results showing that highest documents published in 2023, cited 329 times and the highest cited document is "Evolutionary Dynamics of Multi-Agent Learning: A Survey" in Journal of "Artificial Intelligence" research cited 164 times, Elsevier has the highest publications. Computer science has the highest publication in terms of research area or subject area (75). Out of 49 countries, USA, England, People republic of China, Australia, Denmark and India (total 113 papers) has the

highest documents in terms of collaboration. The highest occurred terms are Algorithm trading, automated trading, market, Machine learning model, prediction, technical analysis etc.

This paper will be helpful for researchers, marketers and interested parties to get an idea of lacking areas so that more deep research can be done with the help of some primary database tools and secondary data research with more factors and with the help of other database source.

Keywords: Algorithm trading; Artificial Intelligence; Retail-investors; Finance; WOS.

Introduction

Algorithmic Trading, a paradigm shift in financial markets, represents the intersection of advanced technology and financial strategies (Currie et al., 2022). At its core, algorithmic trading involves the use of pre-programmed instructions. to execute trading orders at optimal conditions (Yadav, 2018). In recent years, the integration of Artificial Intelligence (AI) has propelled this practice into new dimensions, revolutionizing how financial markets operate (El) (El Hajj & Hammoud, 2023). Algorithmic Trading, often referred to as algo-trading or black-box trading, refers to the use of computer algorithms to automate the process of trading financial instruments (Pothumsetty, 2020) These algorithms are designed to follow a set of predefined rules and execute trades at speeds and frequencies impossible for human traders (Gerner-Beuerle, 2021). This approach has become increasingly prevalent in global financial markets, shaping the dynamics of trading across various asset classes. The infusion of Artificial Intelligence into algorithmic trading has brought about a transformative era (Mossavar Rahmani, 2023) AI, encompassing machine learning and deep learning, allows algorithms to learn from data, adapt to changing market conditions, and make intelligent decisions without explicit programming (Makhlooqa & Mubarakb, 2024) The integration of AI techniques adds a layer of complexity and adaptability, enabling trading algorithms to analyze vast datasets, identify patterns, and optimize strategies in real- time (Reddy, 2023). The symbiosis of algorithmic trading and AI has far-reaching implications for financial markets, influencing liquidity, price discovery, and overall market efficiency (Verma and Sehgal, 2023). This review seeks to delve into the intricate strategies employed in algorithmic trading, exploring how AI technologies have redefined these strategies. Furthermore, it aims to unravel the impact of algorithmic trading on market dynamics, shedding light on its ethical considerations, regulatory landscape, and the evolving trends that will shape the future of financial markets. As the financial industry continues to embrace technological advancements, understanding the nuances of algorithmic trading and AI becomes imperative for traders, regulators, and stakeholders alike. This review serves as a comprehensive exploration of the strategies employed and the market impact wrought by the amalgamation of algorithmic trading and AI. The transformative potency of artificial intelligence—AI—in the already data-driven financial industry conjures notions of Schumpeterian creative destruction. With AI's promises of scale, speed, access to alternative data, automation, bespoke tools and strategies, and low barrier to entry, the investment industry finds itself at an inflection point (An et al., 2021; Stone et al., 2020; Varian, 2018). Accompanying AI's promises are risks and uncertainties. The academic fields of accounting, finance, and economics

confront complex issues arising from AI's impact on market integrity, portfolio management, micro-, macro-, and meso- behaviour by human and algorithmic agents, social welfare, fairness etc (Dwivedi et al., 2021; Issa et al., 2016; West, 2018; Rozanec et al., 2022). An algorithmic agent or an AI agent in the context of this paper is an AI system autonomously or semi-autonomously transacts based on programmed algorithms. It differs from traditional economic agents—who have always been assumed to be human and whose decisions are influenced by cognitive biases and limited processing capacity. Algorithmic agents operate on data-driven decisions. They can handle complex and multivariate environments. Their trades execute within milliseconds. These features have engendered the potential for new market dynamics and systemic risks not accounted for in traditional economic theories. For instance, algorithms stand to weaken bounded rationality—challenging the long-standing notion of efficient markets. Simultaneously, models themselves can be prone to biases—requiring an update of behavioural theories and stylized facts on market anomalies. Moreover, despite two decades of high- frequency trading (HFT), the systemic, market quality, and microstructure effects of algorithmic interactions remain under-mapped. Negotiating these challenges call for proactive, nimble, and value- conscious regulation. This paper tackles these gaps by delving into the salient concerns in the nexus between AI and financial markets, while advocating a need for reconceptualizing financial theories in a multidisciplinary lens.

Research Gap

Artificial Intelligence (AI) has revolutionized algorithmic (algo) trading by enabling the rapid analysis of vast amounts of data, improving decision-making, and optimizing trading strategies. AI algorithms can identify patterns, predict market trends, and execute trades with high precision and speed, reducing human error and increasing efficiency. In high-frequency trading, AI models are crucial for capturing microsecond-level opportunities. However, while institutional traders benefit from sophisticated AI- driven systems, retail investors often face a research gap. Many retail investors lack access to advanced AI tools and data analytics, which limits their ability to compete effectively in the fast-paced markets. This disparity can lead to imbalances in market participation, as retail investors may not have the same insights into market trends, pricing, and risks. Addressing this research gap by providing retail investors with better AI-driven tools and education could democratize access to market opportunities and lead to more balanced trading environments.

So as to do an insightful study based on algo trading the research question framed is to identify the trends and patterns in AI and algorithmic trading for retail investors in last 3 decades, with respect to publication volume, citation impact, and collaborative networks among authors and institutions.

Thus, this paper is an analysis of WOS research papers. There is a lack of research in this topic, availability of data can be helpful for many stakeholders and researchers.

Search terms "Algorithm trading" OR "Automated Trading" OR "Black-Box trading" OR "Algo-trading" AND "Artificial Intelligence" OR "Role of Artificial Intelligence in Algo Trading" AND "Retail-investors" and limiting it to article, proceeding paper, and review article was the main

analysis process.

AI algorithms analyze vast amounts of market data in real-time, identify patterns, and execute trades automatically based on predefined strategies, all within milliseconds. This capability gives institutional investors a competitive edge, allowing them to capitalize on small price movements that are often undetectable to human traders.

Literature review

A Historical Overview Algorithmic Trading and AI

The history of algorithmic trading and artificial intelligence (AI) is a fascinating journey through the evolution of financial markets, technological advancements, and the transformative impact of automation on trading strategies (Azeema et al., 2023). This comprehensive review delves into the key milestones, technological breakthroughs, and paradigm shifts that have shaped the landscape of algorithmic trading and AI. Algorithmic trading traces its roots back to the mid-20th century, with early attempts to automate trading strategies. In the 1950s and 1960s, mainframe computers facilitated the development of basic algorithms for executing trades. However, it wasn't until the 1980s that algorithmic trading gained momentum, thanks to advancements in computing power and the emergence of electronic exchanges. The advent of electronic exchanges in the 1980s marked a pivotal moment in the history of algorithmic trading. The shift from traditional floor trading to electronic platforms paved the way for faster order execution and increased liquidity. Algorithms were employed to optimize trade execution, leveraging speed and efficiency. The 1990s witnessed a surge in technological advancements, including the development of sophisticated algorithms and the widespread adoption of quantitative trading strategies (Nuti et al., 2011) The use of historical data and mathematical models became integral to algorithmic trading. The introduction of decimalization in the early 2000s further fueled the need for advanced algorithms to navigate smaller price increments. The 2000s saw the rise of high-frequency trading, a subset of algorithmic trading characterized by ultra- fast execution speeds and high order-to-trade ratios. HFT firms capitalized on colocation services and direct market access to gain a competitive edge (Zaharudin et al., 2022). This era witnessed both praise for improved market liquidity and concerns about market fragmentation and systemic risks. The last decade has been marked by the integration of machine learning techniques into algorithmic trading strategies. Machine learning algorithms, including neural networks and deep learning, have enabled traders to analyse vast datasets, identify patterns, and make more informed decisions. This era also saw the emergence of AI-driven trading platforms and robo-advisors. As we move into the present and future, the use of AI in trading continues to evolve. Quantum computing, natural language processing, and reinforcement learning are poised to redefine trading strategies. Ethical considerations, algorithmic bias, and regulatory frameworks are gaining prominence as the industry grapples with responsible AI applications (Du & Xie, 2021). In conclusion, the history of algorithmic trading and AI reflects a continuous quest for efficiency, speed, and precision in financial markets. From early computational experiments to the machine learning revolution, each

era has contributed to the ever- evolving landscape of algorithmic trading, shaping the way financial markets operate today (Swathi and Pahuja, 2024).

AI and Finance

The genesis of artificial intelligence in finance harkens back to computational experimentations of the 1980s: the erstwhile unconventional trading strategies of DE Shaw based on what is considered in the modern day machine learning techniques (A Conversation with David Shaw - ACM Queue, n.d.; Biais & Woolley, 2012). Quantitative analysis-based investment, spearheaded by Goldman Sachs in the following decade, preceded the main streamization of statistical arbitrage, pairs trading, and market making algorithms. The 1990s network infrastructure boom and post-dot-com-crash discount prices enabled low-latency network access (Matzkevich & Abramson, 1995). Meanwhile, hardware got stronger and cheaper—as Moore's Law predicted—making possible order initiation and trade execution within milliseconds. This ushered in the age of High Frequency Trading—HFT. A decade later, apprehensions arose concerning market stability, fragmentation, and systemic risk—typified by the HFT-induced flash crash of the Dow in 2010. Since then, 4 AI has permeated into various avenues of the investment management industry (Chui et al., 2018; Jacobides et al., 2021). In the decade since, AI has diffused into practically all branches of the financial industry. Consultancy firms expanded pre- existing AI usage for predictive analytics as part of their strategic advice to clients—in line with conclusions of Chuang et al., (2021). As large datasets became more available, investment tools integrated them to optimize portfolios—something an ordinary investor cannot do with access to generic home PCs or workstations. Corporate risk managers followed suit—employing AI to identify patterns signaling market disruptions. Fraud detection systems used AI to recognize anomalous transactions suggestive of illicit activities. Among the salient literature, worth noting in this regard is the Flash Boys book by M. Lewis—a seminal expose on how low latency and market fragmentation can be exploited for repeated systematic profit which AI agents specialize on. The author was among the first to voice concerns around fairness, transparency, and systemic risks of these practices. On the other hand, a study by SEC economists offer a more contentious empirical evidence on the role of HFTs—hailing them as liquidity providers and originators of price discovery. The downside—high volatility—is presumed to be an acceptable tradeoff. Nonetheless, the paper shows the challenges of isolating and measuring the impact of algorithmic trading. Some literature has also started investigating AI's potentials to democratize investing (Montes & Goertzel, 2019). For instance, "The AI Book" optimistically discusses AI lowering costs and barriers for retail investors to access sophisticated tools and alternative datasets previously exclusive to institutional investors (Bartoletti, 2020). Others adopt a more critical position. For example, Coeckelbergh (2021) argue that AI concentrates power in big tech firms. This can disadvantage marginalized groups without digital access. Significant research gaps persist around detailing AI's long-term influences on retail trading behavior, market dynamics, and cycles of booms and busts.

Objectives

1. To Identify key publication trends, research directions and growth areas in AI and

algorithmic trading for retail investors.

2. To Identify thematic clusters, collaborations between authors, institutions, or geographic distribution.
3. To analyze the most influential papers, authors, journals, citations and impact metrics.
4. To examine the paper's citation network, co-citation, and bibliographic coupling.

Research Methodology

A cmprehensive information on "Algorithm trading" OR "Automated Trading" OR "Black-Box trading" OR "Algo- trading" AND "Artificial Intelligence" OR "Role of Artificial Intelligence in Algo Trading" AND "Retail-investors" associated study is gotten via 2 successive process. The very first stage includes identification of all appropriate documents or publications from 1991-2024 in the WOS database by including a wide variety of Algorithm Trading's relevant keyword phrases: "Automated Trading" OR "Black-Box trading" OR "Algo- trading" etc. This paper implemented literary works evaluation actions recommended by (Keathley-Herring et al., 2016). This treatment aided to manage the compulsory features of the chief lessons and categorize the feasible search keywords. The keyword search included title, abstract and also keywords, as well the investigated duration was readied to take in papers available amid 1991-2024. Papers in English language just were thought about for the evaluation procedure. Presented search query suggested that research study is expanding tremendously. The 2nd phase involves supplementing the presented WOS database with the help of Vosviewer, the process of getting the appropriate information is facilitated by this software program. As necessary, the screening procedure led to a distinct data source of records. After an extensive evaluation complied with technique to literature evaluation. Extra details sources are from available websites and also some more research study documents. The application of contemparary bibliometric strategies lowers the previously mentioned limitations and also offer a reliable way to manage considerable collections of scientific literature (Wang et al., 2020). So far the current research studies have been overlooking so many realities, leading to comparable results and conclusion in this research and also absence of expertise in less-discovered domains (Hossain, 2020), As a result, a cutting-edge quantitative bibliometric evaluation, is performed to supplement the existing research study as well as examine the state-of-the-art on Algorithm trading by cutting-edge software program bibliometric tools such as VOSviewer. PRISMA framework was also used which is an interesting method to remove some irrelevant papers, duplicates were removed and some articles were removed which was not important for this research.

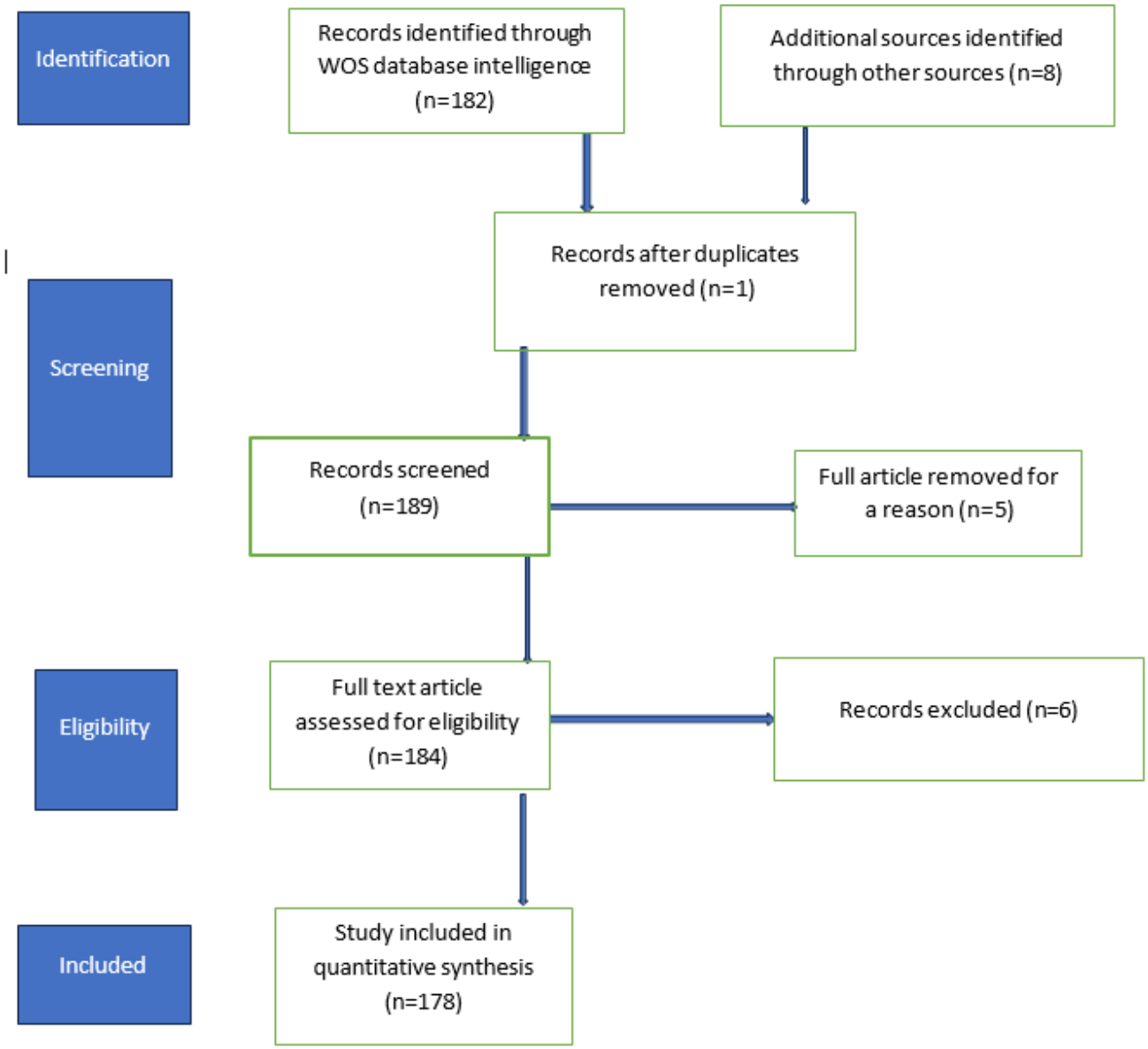


Figure 1: PRISMA Framework (Preferred items for systematic Review and Meta-Analysis PRISMA)

(From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and MetaAnalyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097, http://www.prisma-statement.org/)

Data and Key Findings

Summarised files used in this research study exist as: A total of 182 files composed by various writers and published in numerous journals were made use of in this research study. As per the figure 2 the highest documents published in 2023, cited 329 times and the highest cited document is "Evolutionary Dynamics of Multi-Agent Learning: A Survey" in journal

of Artificial Intelligence research cited 164 times. After limiting the papers to research, articles, language English and after removing some irrelevant papers the final output was 178, which was cited 2,489 times with h-index 27 (figure 2.1). Elsevier, has the highest publications followed by Springer, Taylor, Wiley, IEEE and Sage (figure 3). In figure 4 top 10 research areas as per the publications are Computer science (75), Business Economics (63), Engineering (29), Operations research management science (26), Mathematics (21), Mathematical methods in social science (8), Automation control system, Information science Library science, Telecommunication (5), Environment science ecology (40). though the main focus was on "Computer Science Artificial Intelligence". The search engine was limited to language English only and then the final paper were 178, out of them highest documents published by Borch, Casey.

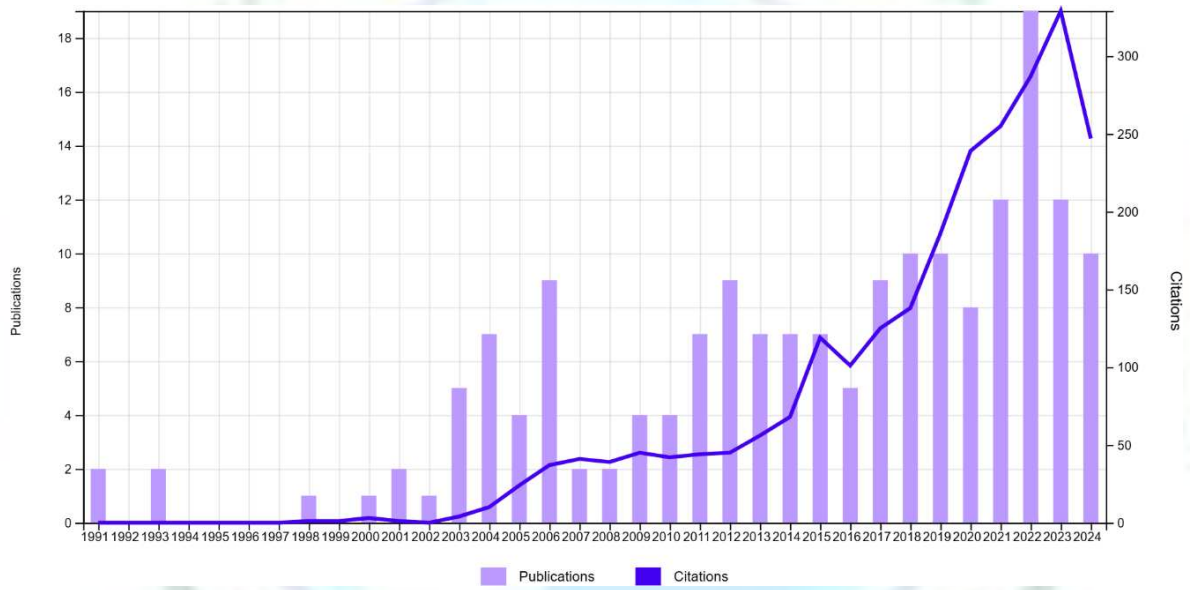
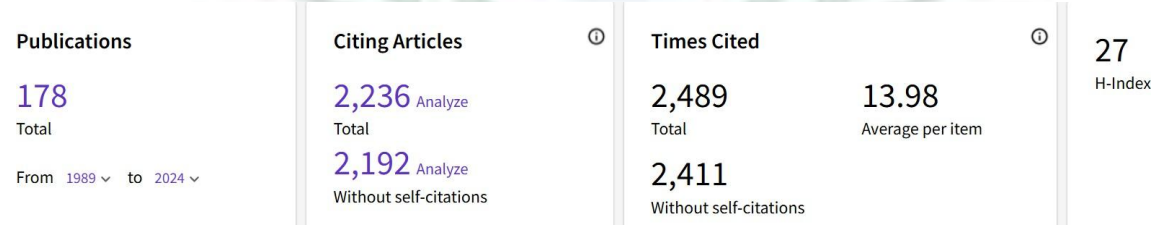


Figure 2: Publication and Citations, Source: WOS database



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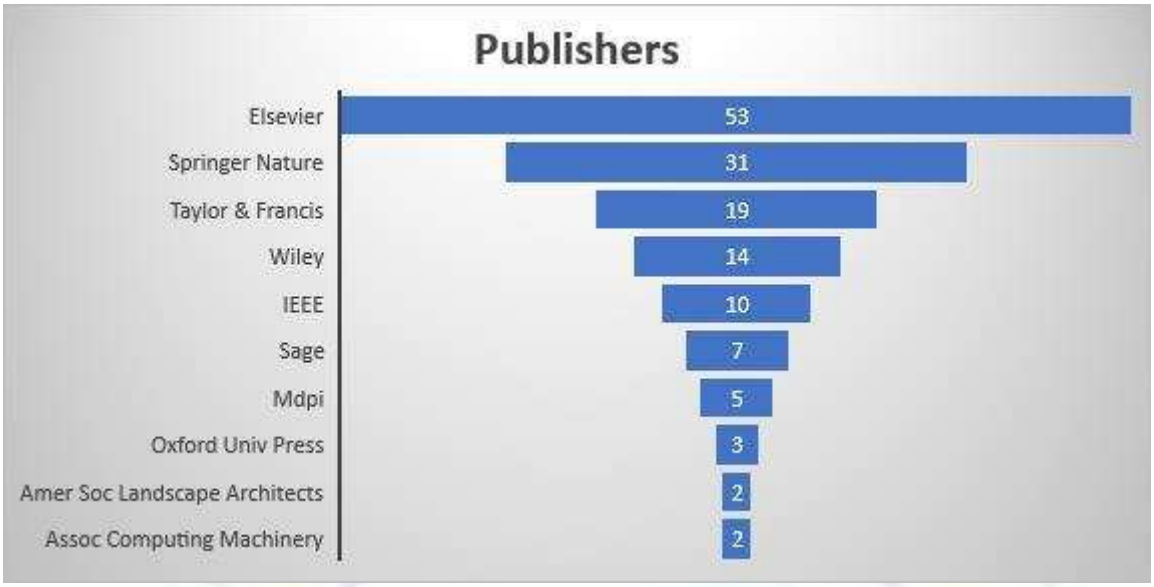


Figure 3: Highest publishers, Source: Authors composition

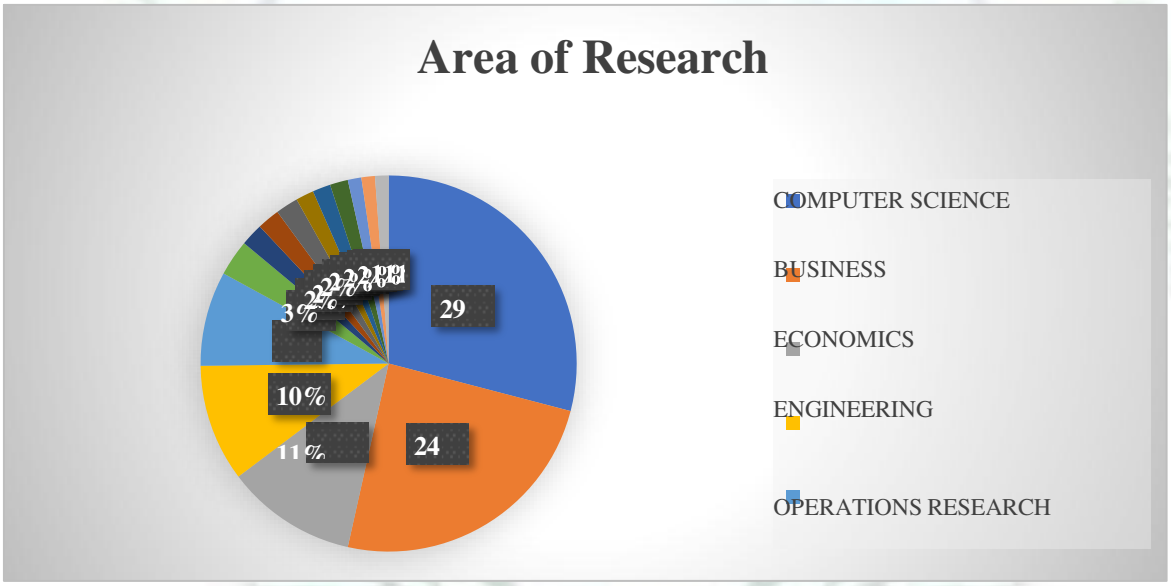


Figure 4: Area of Research, Source: Authors composition

Co-Authorship Analysis of Countries

As per map of countries co-authorship (international collaboration), there are 4 clusters with 9 items and 16 links. Out of 49 countries, USA, England, People republic of China, Australia, Denmark and India (total 113 papers) has the highest documents in terms of collaboration. Further, figure 4.1 shows the country wise publications as per which USA, PRC, England, Australia, Denmark has the highest papers.

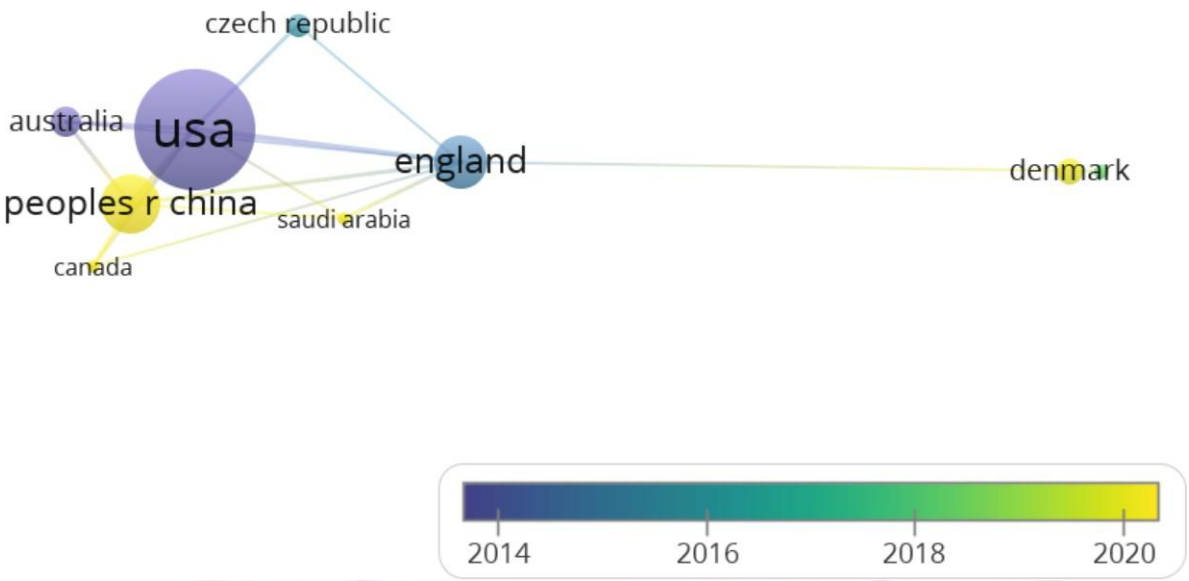


Figure 5: Co-Authorship Analysis of Countries, Source: VOSviewer

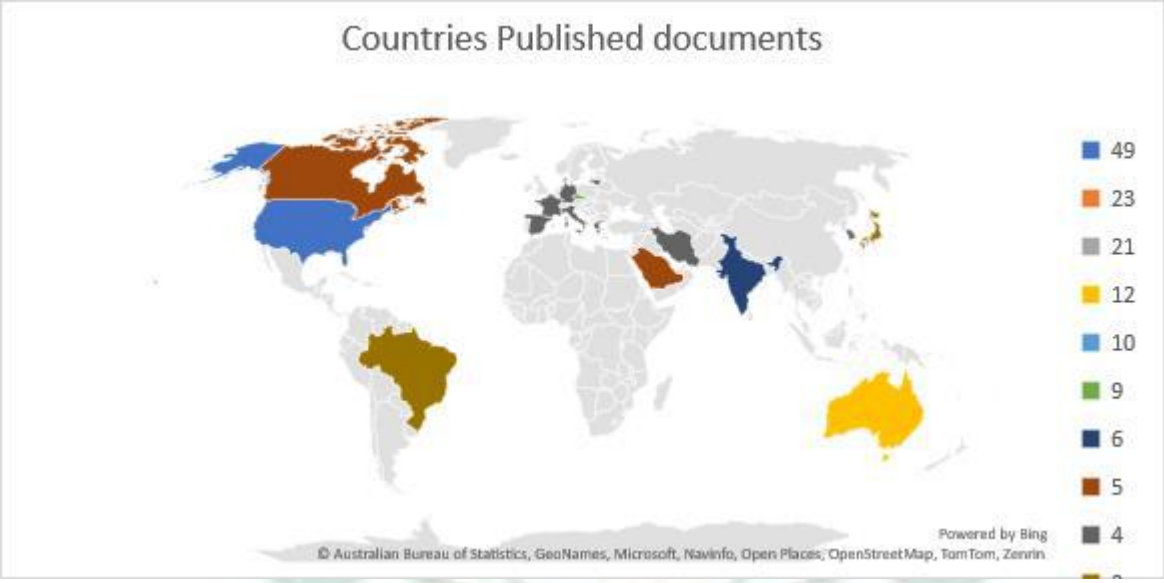
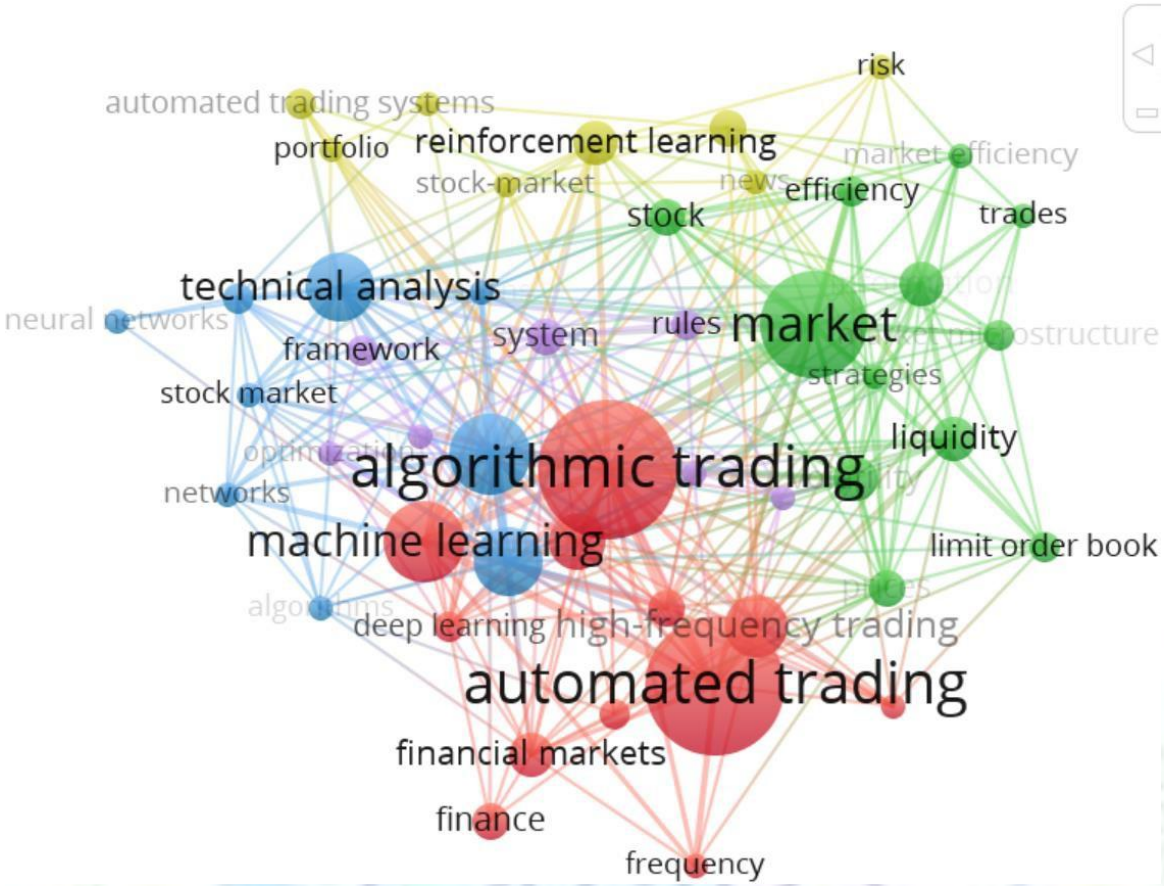


Figure 5.1: Countries publications, Source: Authors Composition

Co-occurance of Keywords

As per the figure co-occurance of keywords was done with the help of Vosviewer, 5 clusters with 48 items and 304 links were made. First and second cluster consist of 12 items each, third cluster consist of 9 items, forth cluster consist of 8 items and fifth cluster is having items. For further analysis below table 1 is given which showing the keyword, its occurance and strength.



The highest occurred terms are Algorithm trading, automated trading, market, Machine learning model, prediction, technical analysis etc.

Keyword	Occurrences	Total link strength
algorithmic trading	22	34
automated trading	22	24
market	17	21
machine learning	13	19
model	13	18
prediction	11	14
volatility	7	14
markets	9	11
technical analysis	11	11
high-frequency trading	10	10
liquidity	7	10
financial markets	7	9
reinforcement learning	7	8
information	7	7

Table 1: Keyword occurance, Source: Vosviewer

Implications

- Guidance for Future Research: The analysis highlights gaps and underexplored areas, offering direction for future studies. Researchers can focus on emerging themes or overlooked topics that need more investigation.
- Influence on Policy and Practice: Insights into the most influential studies and trends can help policymakers and industry practitioners align their strategies with current research, particularly in technology-driven fields like AI and algorithmic trading.
- Enhanced Collaboration Opportunities: By identifying key authors, institutions, and countries leading in this field, the analysis can foster potential partnerships, collaborations, or networks for academic and industry stakeholders.

Conclusion

Artificial Intelligence (AI) in algorithmic (algo) trading has significantly reshaped financial markets, allowing for the automation of trading strategies, predictive analytics, and real-time decision-making. AI-driven models can process vast amounts of data in seconds, enabling institutional traders to identify patterns, forecast market movements, and execute trades at speeds far beyond human capability. This advancement has provided a competitive edge to large financial institutions, optimizing trade execution and risk management. AI plays a significant role in algorithmic trading by enabling more efficient, faster, and data-driven decision-making processes. For retail investors, AI-driven trading can create more efficient and liquid markets. Papers are available on this topic but less papers are considering all three factors, AI, Algo-trading and Retail-Investors, neither a vast, scientific research is done. By, keeping that gap in mind this paper is an analysis of WOS research papers. Terms were searched on WOS -"Algorithm trading" OR "Automated Trading" OR "Black-Box trading" OR "Algo- trading" AND "Artificial Intelligence" OR "Role of Artificial Intelligence in Algo Trading" AND "Retail- investors", 181 papers were generated, after limiting the papers to article, proceeding paper, review article and language English the final papers were 178, the highest documents published in 2023, cited 329 times and the highest cited document is “Evolutionary Dynamics of Multi-Agent Learning: A Survey” in journal of Artificial Intelligence research cited 164 times. After limiting the papers to research, articles and language English the final output was 178, which was cited 2,489 times with h- index 27. Elsevier, has the highest publications followed by Springer, Taylor, Wiley, IEEE and Sage. Top 10 research areas as per the publications are Computer science, Business Economics, Engineering, Operations research management science, Mathematics, Mathematical methods in social science, Automation control system, Information science Library science, Telecommunication, Environment science ecology. though the main focus was on “Computer Science Artificial Intelligence”. The final papers were 178, out of them highest documents published by Borch. As per map of countries co- authorship (international collaboration), out of 49 countries, USA, England, People republic of China, Australia, Denmark and India (total 113 papers) has the highest documents in terms of collaboration. Further, country wise publication shows USA, PRC, England, Australia, Denmark has the highest papers. As per the figure co-occurance of keywords was done with the help of

Vosviewer, the highest occurred terms are Algorithm trading, automated trading, market, Machine learning model, prediction, and technical analysis .

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